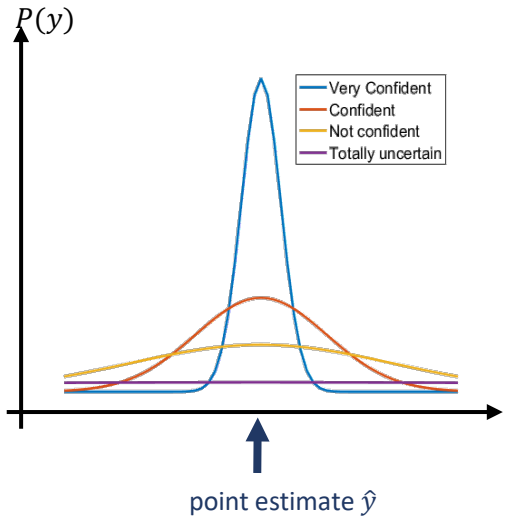
Recap - Uncertainty Estimation



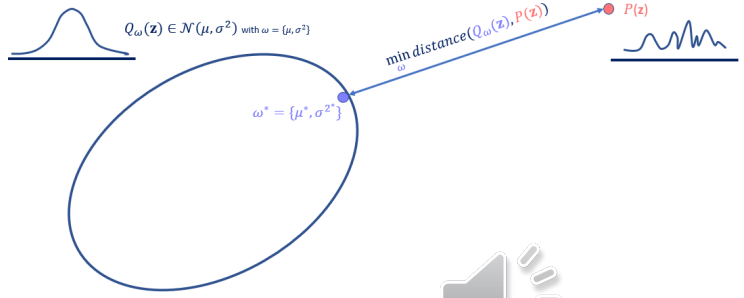
Aleatoric Uncertainty (irreducible uncertainty) → Maximum-Likelihood Estimation of Noise

Epistemic Uncertainty (decreases with more data)

Bayesian Modeling $P(\boldsymbol{\theta}|D) = \frac{P(D|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(D)}$

$$\mathrm{P}(\mathcal{D}) = \int P(\mathcal{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})d\boldsymbol{\theta} = \int\int \ldots \int \ldots d\theta_1 d\theta_2 \ldots$$

Approximation Techniques



Variational Approximation $\min_{\boldsymbol{\omega}} D_{KL}\left(Q_{\boldsymbol{\omega}}(\mathbf{z}) \parallel P(\mathbf{z}|\mathbf{x})\right)$

Bayesian Neural Network

Big picture

- Simple to formulate and understand
  - *e.g.,* Gaussian priors on individual weights
  - *e.g.,* likelihood functions such as softmax or Gaussian

- Potentially difficult (and computationally expensive) to infer
  - Make many approximations/restrictions (Binarization of parameters and/or inputs, Bernouli distributions, Gaussian Distributions, Sampling, etc.)

[Brief but good overview of the old methods in Gal's "Uncertainty in Deep Learning" Chapter 2.2.1]

What if I don’t want to change the *training* procedure (too much)?



Modern Approaches: Simple&Effective Techniques

- Why?
  - it's simple to use (easy for practitioners to use and understand)
  - it works with pretrained models
  - does not change the heavily optimized architecture
  - does not change the heavily optimized loss function
  - does not change the heavily optimized optimization procedures
  - no (or low number of) additional hyper-parameter to tune
  - (potentially) more efficient training
  - scale well to large data

